Fast Inference from Transformers via Speculative Decoding

Abstract
Inference from large autoregressive models like Transformers is notoriously slow because generation is strictly sequential: generating K tokens requires K serial forward passes. We introduce Speculative Decoding, an algorithmic framework that speeds up decoding without altering the model's output distribution. A smaller, faster draft model generates candidate tokens speculatively, and the larger target model evaluates them in parallel in a single forward pass.

1. Introduction
Autoregressive sequence generation models compute output probabilities token-by-token. For modern Large Language Models (LLMs) running on modern accelerators (GPUs/TPUs), the latency of executing a forward pass is dominated by memory bandwidth (reading model weights from High Bandwidth Memory into on-chip SRAM) rather than arithmetic compute. Consequently, generating a single token requires almost the same wall-clock time as computing several tokens in parallel.

2. Methodology & Algorithm
Speculative Decoding leverages two models:
1. A draft model M_{draft} (e.g., 100M parameters) which is computationally cheap and memory-efficient.
2. A target model M_{target} (e.g., 70B parameters) whose exact probability distribution we wish to preserve.

At each step:
- M_{draft} autoregressively predicts a draft sequence of length $K = \gamma$ (typically $\gamma \in [3, 6]$).
- M_{target} executes a single forward pass over all K proposed tokens in parallel.
- A modified rejection sampling scheme is applied to determine how many prefix tokens are accepted:
  $$\alpha = \min\left(1, \frac{P_{target}(x)}{P_{draft}(x)}\right)$$
- If a token is rejected, a replacement token is sampled from the adjusted distribution $(P_{target}(x) - P_{draft}(x))^+$, and the remaining proposed tokens are discarded.

3. Empirical Evaluation & Benchmarks
We evaluate Speculative Decoding across T5-XXL, Chinchilla, and PaLM architectures on standard benchmarks (WMT, CNN/DailyMail, HumanEval).
- Speedup: Observed wall-clock latency reduction ranges from 2.0x to 2.8x depending on domain entropy and draft model alignment.
- Output Equivalence: The output distribution is mathematically identical to standard autoregressive sampling from M_{target} (sampling difference $\equiv 0$).
- Acceptance Rate: Mean acceptance rate β per token ranges between 65% and 82%.

4. Limitations & Trade-offs
- Memory Footprint: Requires hosting both draft and target models in accelerator memory simultaneously.
- Draft Model Training: Obtaining high acceptance rates requires the draft model vocabulary and tokenizer to match the target model.
- Low-latency / High-concurrency: Under high batch-size conditions where the GPU is compute-bound rather than memory-bandwidth-bound, the relative speedup diminishes.